Lucas A. **Saavedra**

Computer Engineer · AI Researcher

Buenos Aires, Argentina

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Experience

Accenture

Buenos Aires, Argentina

Senior Full-Stack AI Developer — *May 2025 - PRESENT*

- GenAI Studio team: LLM-based multi-agent systems (LangGraph, FastAPI, Azure OpenAI, Elasticsearch) for automated code migration between languages, data-quality testing, and AI-assisted accounting reconciliation.
- Social media analytics platform: Neo4j knowledge graph, Spring Boot batch classification pipeline, and a LangGraph conversational agent exposed through a Next.js chat interface, including an MCP server to expose the graph as tools for the agent.
- Real-time computer vision systems (YOLO, OpenCV, TensorFlow) for vehicle detection, customer profiling, and dataset preparation for gas station clients.
- Full-stack development with Next.js/React and enterprise Angular applications, deployed with Docker and Azure Pipelines.
- Research on Small Language Models (SLMs) and development of agents to automate the Software Development Life Cycle (SDLC) through Spec-Driven Development.
- **Technologies used: Python, Large Language Models (LLMs), LangGraph, FastAPI, Azure OpenAI, ElasticSearch, Neo4j, Spring Boot, Next.js, React, Angular, Docker, Azure Pipelines, Computer Vision (YOLO, OpenCV, TensorFlow).**

BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

Research Assistant — *Jan. 2022 - PRESENT*

- Implemented Deep Learning models for static and dynamic analysis of the nicotinic acetylcholine receptor (nAChR), with orientation to graph theory, recurrent networks and geometric computing.
- Analyzed distribution and density of nicotinic acetylcholine receptors in the plasma membrane using super-resolution microscopy (STED, STORM), with advanced analytics implemented in MATLAB based on graph theory and computational geometry.
- Developed an efficient and autonomous algorithm based on supervised Graph Neural Networks (GNNs) in TensorFlow to detect and quantify dynamic particle clustering in single-molecule microscopy (SMLM) data.
- Applied AlphaFold3-equivalent models to study protein-ligand structures (nAChR and cholesterol), combined with molecular dynamics (MD) simulations using GROMACS on high-performance computing (HPC) platforms with SLURM.
- Developed WadTCN, a deep learning architecture for 2D single-particle diffusion analysis (anomalous exponent and diffusion coefficient estimation, model classification, trajectory segmentation), applied to nAChR trajectories in live cells.
- Analyzed two-colour sub-millisecond MINFLUX single-molecule tracking data of nAChR and cholesterol in live cells, with trajectory segmentation and diffusive-regime classification.
- Design and programming in C++ of electronic circuits with Arduino to control piezo motors.
- **Technologies used: Python, Java, C++, TensorFlow, PyTorch, Pandas, NumPy, Docker, MongoDB, MATLAB, GROMACS, SLURM, Arduino.**

Data Science and AI Laboratory, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

Research Assistant — *Jan. 2025 - Jan. 2026*

- Supervisor: Prof. Antonio Quintero-Rincón.
- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning—specifically, hierarchical variational autoencoders—to detect specific cognitive activity with PyTorch and MNE-Python.

Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

Assistant Professor — *Mar. 2023 - June 2025*

- ”Introduction to Data Science”: Artificial Intelligence concepts, Neural Networks, Scientific Method.
- ”Data-Oriented Programming”: Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- ”Discrete Mathematics”: Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

UpWork

Freelancer — *Dec. 2023 - May 2025*

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2 Linux instances.
- YOLO training to identify basketball players in video matches from a European basketball league, with a SQL database hosted in Supabase.
- GUIs developed with built-in Napari plugins to gather information from laser spot images, combined with decision tree classifiers in scikit-learn.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium, Napari, scikit-learn.**

Deloitte

Buenos Aires, Argentina

Back-End Developer — *Jun. 2021 - Jun. 2022*

- Developed Codex Gigas, a malware ”DNA” search engine that identifies patterns and characteristics for malware hunting usign REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

Deloitte *Buenos Aires, Argentina*

Cyber Risk Advisory Intern *Jan. 2021 - Jun. 2021*

- Offensive/Defensive Security and Threat Intelligence basic concepts.

Skills

DevOps	AWS, Docker, CircleCI
Technologies	Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis, Selenium, Docker, YOLO, Langgraph, Langchain, Azure OpenAI
Programming	Python, JAVA, C, C++, R, LaTeX, MATLAB

Education

Pontificia Universidad Católica Argentina (UCA) *Buenos Aires, Argentina*

Computer Engineering *2018 - 2023*

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)
- **GPA:** 8.02/10.

Publications and conferences

2022 **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences

2023 **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium *Punta del Este, Uruguay*

2024 **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale

2024 **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0

2025 **Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis**, Journal of Physics: Photonics

2025 **Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking**, Nature Communications

2025 **Quantitative evaluation of methods to analyze motion changes in single-particle experiments**, Nature Communications

2025 **Environmental enrichment delays the onset of cognitive decline in a transgenic mouse model of Alzheimer disease**, XL Annual Meeting SAN

2026 **Machine Learning in Single-Molecule Tracking Analysis of Superresolution Optical Microscopy Data**, Cells

2026 **A self-supervised reconstruction framework for segment-level detection of task-induced deviations in multi-channel EEG**, PLOS ONE (Under review)

Certifications and Courses

2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
2020 **CS50's Introduction to Artificial Intelligence with Python**, Harvard, edX
2021 **Machine Learning**, Stanford Online, Coursera
2021 **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
2021 **R Programming**, Johns Hopkins University, Coursera
2021 **Docker for the Absolute Beginner - Hands On - DevOps**, Udemy
2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
2022 **AWS Certified Cloud Practitioner**,
2022 **Data Science**, Coderhouse
2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
2025 **Introduction to Statistics**, Stanford Online, Coursera
2025 **Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)**,
2025 **Bioinformatics; Learn Docking & Mol Dynamics Simulation**, Udemy
2025 **Basic to Advanced: Retrieval-Augmented Generation (RAG)**, Udemy
2025 **IELTS Score: 7.5**,
2026 **Golang for Beginners**, Udemy
2026 **Kubernetes for the Absolute Beginners - Hands-on**, Udemy